



PREDICTING STUDENT STRESS LEVELS USING PSYCHOLOGICAL, PHYSIOLOGICAL, AND ACADEMIC FACTORS

Group 09



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Abstract

This study analyzed data from students across the nation collected through a comprehensive survey to identify key factors influencing stress levels — categorized as No Stress, Eustress, and Distress. After cleaning the data and removing duplicates, cluster and outlier analyses were performed to explore patterns among psychological, lifestyle, and academic variables. Several predictive models, including Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost, were compared. Among them, the Random Forest model achieved the best performance in detecting distressed students. The results reveal that blood pressure, sleep quality, teacher–student relationship, and academic performance are the strongest predictors of distress, suggesting that targeted interventions in these areas could help reduce student stress and improve overall well-being.

Introduction

Effective stress management is crucial for maintaining both physical and mental health. Among students, unmanaged stress has become a serious public-health concern due to rising academic pressure, social expectations, and lifestyle imbalance. When stress becomes chronic or overwhelming, it can trigger anxiety, depression, sleep disturbances, and even suicidal thoughts. Research by the World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) shows that suicide is one of the leading causes of death among people aged 15–29, and that early identification and treatment of high-stress individuals can prevent most of these outcomes.

By recognizing students at risk early, schools and universities can offer counseling, promote healthy coping skills, and strengthen support systems—such as peer mentoring and positive teacher–student relationships. These interventions not only protect individual well-being but also build safer, more cohesive communities. Evidence suggests that students who receive timely psychological or academic support are less likely to engage in violent or self-destructive behavior, contributing to long-term social stability.

This project aims to analyze nationwide survey data to determine which psychological, physiological, and lifestyle factors most strongly predict stress levels among students. The findings can guide educational institutions and policymakers in developing data-driven prevention strategies that improve student welfare and reduce community-level mental-health risks.

Description of the question

Student stress is a multidimensional issue influenced by psychological, lifestyle, and academic factors. While mild stress (eustress) can enhance motivation, excessive stress (distress) impairs concentration, physical health, and social relationships. Educational institutions often lack data-driven methods to identify students at risk before distress escalates into serious mental health or behavioral issues.

This study seeks to answer three key questions:

1. Which factors most strongly contribute to student distress?
2. Can students be grouped into meaningful stress-profile clusters based on measurable features?
3. Which predictive model most accurately classifies stress levels (No Stress, Eustress, Distress)?

By combining statistical learning techniques with interpretable results, the analysis aims to support timely interventions that enhance student well-being and, ultimately, foster a healthier and safer society.

Data Pre-processing

Before conducting the analysis, the dataset was carefully checked for quality and consistency. The inspection revealed no missing values, no duplicate records, and no data-entry errors. Therefore, the dataset was considered clean and ready for analysis without requiring additional imputation or correction steps.

This ensured that all subsequent results were based on accurate, complete information and that any detected patterns reflected genuine relationships rather than data inconsistencies.

Exploratory Data Analysis Results

Correlation Analysis

To explore how psychological, physiological and lifestyle factors interact, a Phi-K correlation analysis was carried out. This method is well suited for mixed data, as it can capture relationships among both numeric and categorical variables.

The results showed several strong and meaningful links, Stress level was most closely related to blood pressure (0.99), sleep quality (0.98), teacher–student relationship (0.95), and academic performance (0.94). In simple terms, students who reported poor sleep, higher blood pressure, weaker connections with teachers, and lower grades tended to experience higher stress.

There was also a clear connection between anxiety and depression (≈ 0.70), which is expected since both describe overlapping emotional states. On the other hand, self-esteem showed the opposite pattern—students with stronger self-esteem generally experienced less stress and anxiety, suggesting greater emotional resilience.

While some variables were closely related, they were all kept in the dataset. Many of these measures describe different but related aspects of student well-being and removing them could oversimplify the patterns of stress the study aimed to understand.

Outlier analysis

Outlier detection was performed to identify unusual observations that differed significantly from the overall student population. When examined individually, the three main numerical variables anxiety level, depression, and self-esteem showed no extreme univariate outliers. However, a multivariate approach using Mahalanobis distance combined with Factor Analysis of Mixed Data (FAMD) identified 22 multivariate outliers.

These cases represented students with uncommon combinations of psychological and lifestyle patterns rather than data-entry errors. Because they reflected genuine but rare profiles, they were retained in the dataset to preserve the full diversity of student behaviors. Removing them could have reduced the representativeness of the model and limited its ability to detect real-world extreme cases of stress.

Cluster Analysis

To explore whether students could be grouped into distinct stress profiles based on psychological, physiological, and academic factors, a clustering analysis was initially conducted using K-means. The aim was to identify latent patterns in the data that could potentially inform targeted interventions.

The analysis identified preliminary clusters; however, evaluation using the silhouette score indicated a value of 0.35, suggesting only weak separation between the groups. In behavioral datasets such as student stress, where stress levels exist on a continuum rather than forming clearly distinct categories, moderate silhouette scores are common. This result indicated that the clusters were not well-defined or strongly distinct, and therefore could not provide reliable information for predictive modeling.

Given this, the clusters were not used as features in subsequent models. Instead, the focus remained on individual psychological, physiological, social, and academic variables that were shown to have strong predictive relationships with stress levels. This approach ensured that the predictive models were based on robust, interpretable features rather than potentially misleading cluster assignments.

Overall, while the clustering exercise provided an exploratory view of the data, it confirmed that student stress profiles are highly individualized and continuous, rather than forming sharply separated groups, supporting the decision to focus on feature-based modeling.

Important findings from explanatory data analysis

The boxplots for the key psychological variables anxiety, depression, and self-esteem clearly show how these factors differ across the three stress levels. Students in the Distress group reported the highest anxiety and depression and the lowest self-esteem, while those in the No-Stress group showed the opposite pattern. The Eustress group fell in between, suggesting moderate stress that remains manageable and can even help motivation. These patterns highlight the emotional contrast between the three groups.

Since most other variables are categorical, their relationships with stress level were examined using cross-tabulations and Chi-square tests. Results confirmed that nearly all lifestyle and academic factors were significantly related to stress. The strongest associations were seen for sleep quality, career concerns, bullying, teacher–student relationship, and academic performance. Students who slept poorly, felt unsupported by teachers, worried about their future, or performed poorly academically were far more likely to experience distress.

It’s also important to note that the response classes were nearly balanced. Overall, the visual and statistical evidence shows that psychological, academic, and social factors all play major roles in student stress, emphasizing the importance of early detection and supportive interventions

Feature Selection

To identify the most relevant predictors, we used multiple feature selection techniques and compared their results. Based on performance consistency and interpretability, the final model used 11 key features identified primarily through Random Forest feature importance and correlation analysis, which together provided the best predictive balance (F1-score: 0.9045).

The selected features reveal that student stress is best predicted by a combination of sleep quality, psychological well-being (self-esteem, anxiety, depression), social factors (bullying, peer pressure), academic pressures (performance, future career concerns), and basic physiological needs. This optimal subset represents a balanced combination across all measured domains.

Selected features: sleep_quality, self_esteem, bullying, anxiety_level,
academic_performance, depression, blood_pressure, basic_needs,
future_career_concerns, teacher_student_relationship, peer_pressure

Identifying Latent Features & Interacting Terms

A Factor Analysis for Mixed Data (FAMD) was used to uncover hidden relationships among the psychological, physiological, and social variables. The scree plot showed an elbow at two components, which together explained about 65.7% of the total variance, meaning that most information in the dataset can be summarized by two broad factors.

The heatmap of variable contributions revealed that blood pressure, sleep quality, and breathing problems loaded strongly on the first component, representing a General Stress Factor—overall physiological and emotional strain. The second component was influenced mainly by social support and future career concerns, forming a Physiological–Social Factor that reflects external social and environmental pressures.

In simple terms, these two components capture the internal health-related and external social dimensions of student stress, offering a compact view of how different aspects of student life combine to influence well-being.

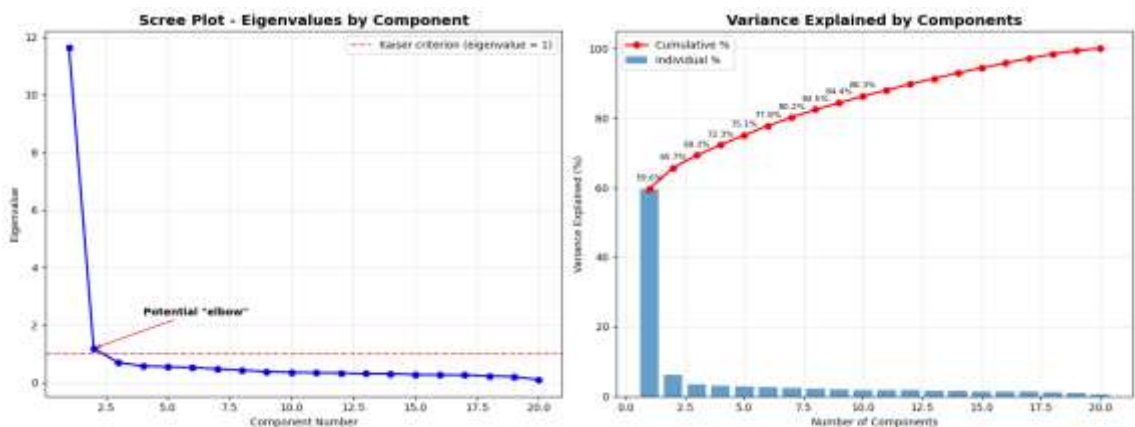


Figure 1:FAMD scree plot

All Models performance comparison.

Then to evaluate how different machine learning techniques perform in predicting student stress levels, we tested the following models. These models were selected to balance interpretability with predictive power.

- Random Forest - Handles mixed data types and non-linear relationships well; useful for identifying key stress factors.
- XGBoost - Captures complex patterns with high accuracy and robustness to outliers and missing values.
- LightGBM - Similar to XGBoost but faster and more efficient for model experimentation.

- Logistic Regression - Simple and interpretable baseline for understanding predictor effects on stress levels.
- Support Vector Machine - Models non-linear relationships effectively, suitable for multi-factor psychological data.
- Neural Network - Learns complex patterns but needs careful tuning due to limited sample size.

Model	Accuracy		F1 Score			
	Train	Test	Train		Test	
			Weighted Avg F1	Distress F1	Weighted Avg F1	Distress F1
Random Forest	1.0000	0.9045	1.0000	1.0000	0.9045	0.8933
XGBoost	1.0000	0.8727	0.9364	0.9384	0.8678	0.8742
LightGBM	0.9398	0.8682	0.9398	0.9396	0.8677	0.8609
Logistic Regression	0.9057	0.8864	0.8848	0.8724	0.8878	0.8571
SVM	1.0000	0.8727	1.0000	1.0000	0.8717	0.8874
Neural Network	0.8864	0.8818	0.8864	0.8763	0.8817	0.8667

Table 1: All models’ performance comparison

After evaluating all models, the Random Forest achieved the highest overall performance across all key metrics, including accuracy, precision, recall, and F1 score. It demonstrated strong generalization ability with a test accuracy of 0.90 and a high F1 score for the high-stress class, outperforming other algorithms such as XGBoost, LightGBM, SVM, and Neural Network. Therefore, we interpret the **Random Forest** model the best performer using feature importance and partial-dependence analyses for the top predictors of **Distress**.

Random Forest Model Performance

Random Forest was used for its strength in handling outliers and multicollinearity while capturing complex feature interactions.

	Test Accuracy	Test F1 Score	Test Distress F1
With all features	0.8955	0.8952	0.8889
With selected features	0.9045	0.9045	0.8933
With latent features	0.8733	0.8730	0.8821

Table 2: Accuracy table with important features

Model performance for RF with selected features

Classes	Precision	Recall	F1 Score
No Stress	0.90	0.88	0.89
Eustress	0.93	0.93	0.93
Distress	0.88	0.91	0.89

Table 3: Performance of each class in response

Overall performance: Using selected features slightly improved the model’s accuracy and F1 score compared to using all features (0.9045 vs 0.8955), showing that feature selection helped the model focus on the most important predictors. Using latent features reduced performance, indicating they may not capture stress-related patterns well.

Distress is predicted well, with F1 = 0.89 and recall = 0.91, meaning most high-stress cases are correctly identified.

Feature Selection and Insights

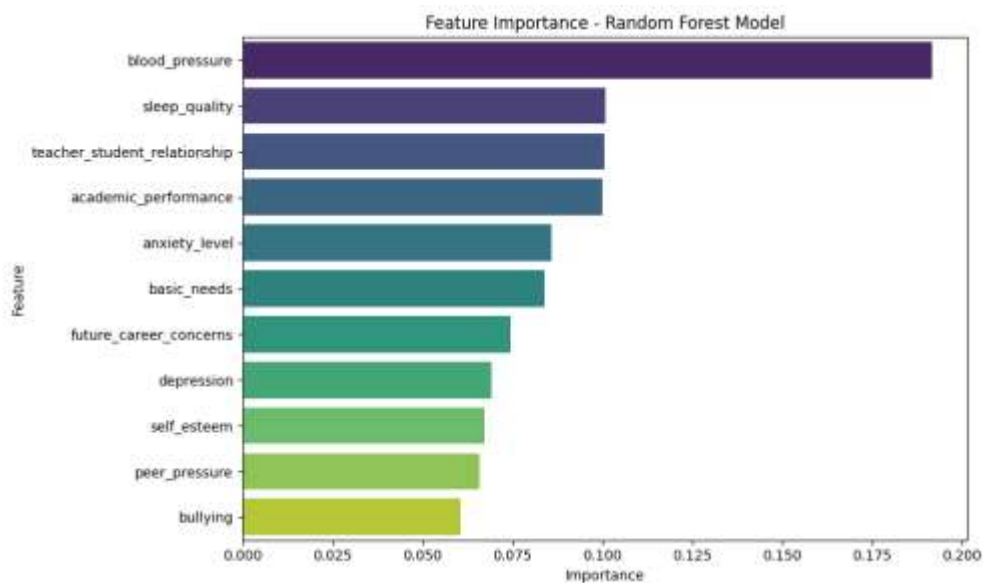
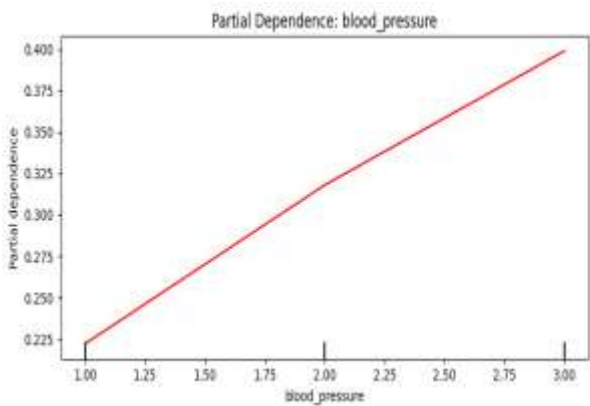


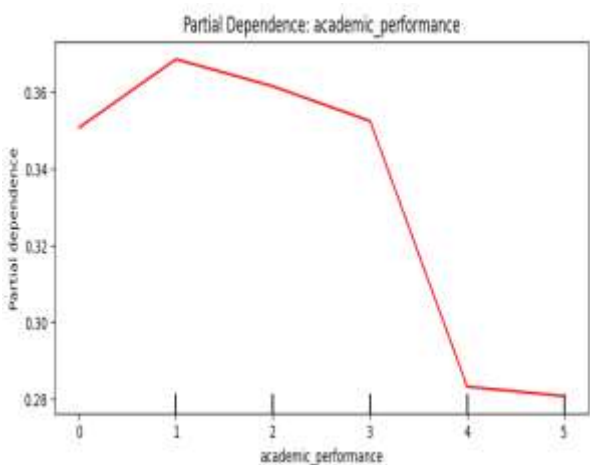
Figure 2: Feature importance plot of random forest model

Blood pressure is the most influential factor in predicting stress levels, followed by sleep quality and teacher-student relationships, emphasizing the role of both physical and mental health. Moderate importance is observed for academic performance, anxiety levels, and basic needs, indicating the influence of social and emotional factors. Meanwhile, future career concerns, depression, self-esteem, and peer pressure have lower but still notable contributions to the model's predictions.

Partial dependency plot analysis (For Distress class)

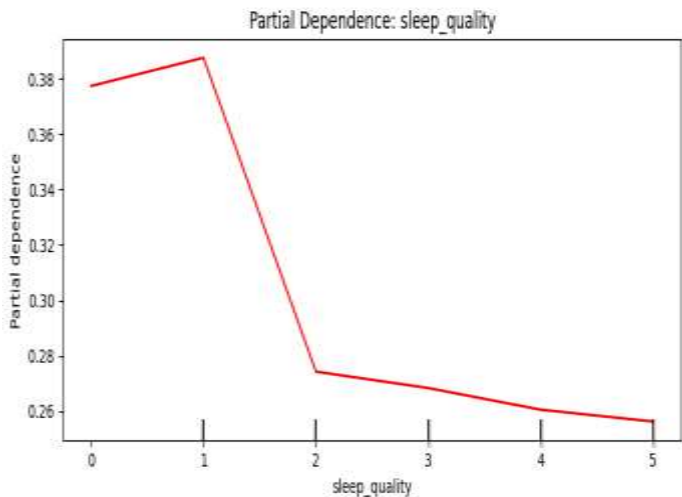


- The plot shows a clear positive relationship between blood pressure and stress. As blood pressure increases from 1.0 to 3.0, the predicted stress level rises steadily from 0.25 to 0.40.
- This indicates that higher blood pressure is strongly associated with a higher probability of stress, which aligns with the known physiological link between the two.



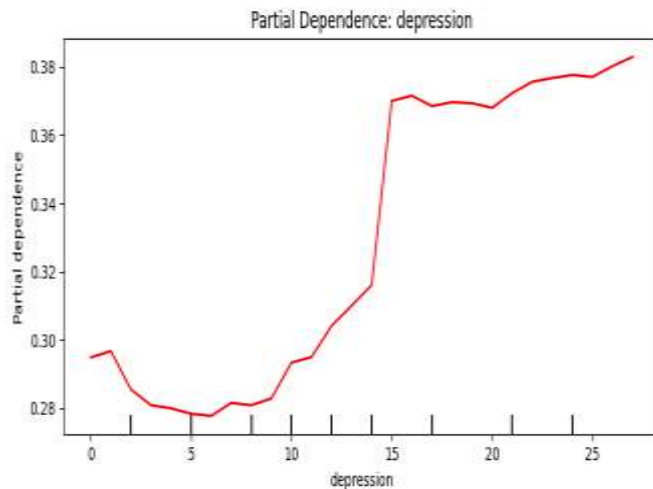
contribute strongly to student stress.

- The partial dependence plot for academic performance shows Students with the lowest academic performance scores have the highest predicted stress while those with the highest scores (around 5) show much lower stress.
- This decline indicates that students who perceive their academic performance positively are less likely to experience Distress, aligning with the idea that poor academic outcomes

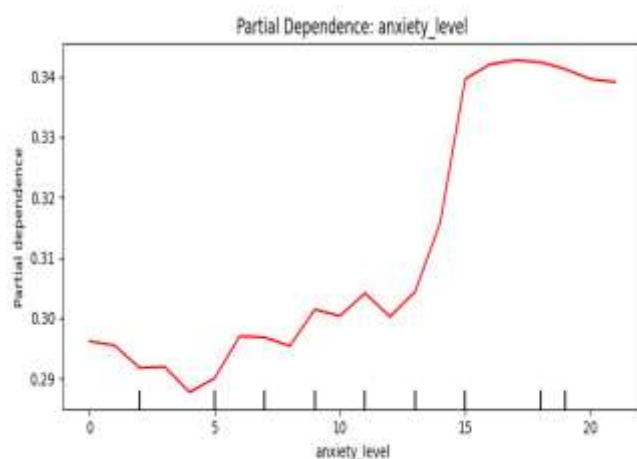


Distress.

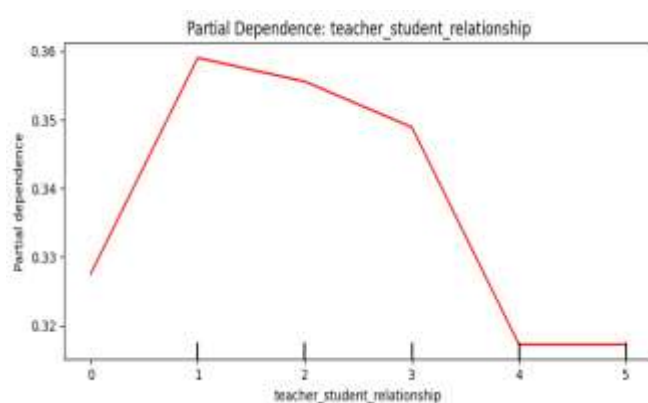
- The partial dependence plot for sleep quality shows as sleep quality improves, the predicted probability of Distress decreases sharply and consistently.
- This suggests that students who report better sleep quality are considerably less likely to experience



- The partial dependence plot for depression reveals low depression scores, the predicted stress begins around 0.28 and increases steadily as depression levels rise, reaching about 0.38 at the highest scores.
- This indicates that students with higher depression symptoms are much more likely to experience Distress.



- The partial dependence plot for anxiety_level demonstrates as anxiety increases, the predicted stress level rises consistently, reaching approximately 0.34 at the highest anxiety values.
- This upward trend confirms that higher anxiety scores are directly associated with a greater probability of Distress.



levels.

- The partial dependence plot for teacher_student_relationship reveals as the quality of the teacher-student relationship improves the predicted stress level shows a decline.
- This indicates that students who report more positive relationships with their teachers are predicted to have lower stress

Issues Encountered and Proposed Solutions

Outliers in the Dataset

- We identified potential outliers using Mahalanobis Distance.
- However, we did not remove them, as there was no strong evidence to justify their exclusion.

Limited Predictive Power of Latent Factors

- We identified two latent components using Factor Analysis for Mixed Data (FAMD).
- However, these components showed low variance explanation and did not improve prediction accuracy.
- We pivoted to using direct feature selection, which yielded superior results.

Mixed Data Types and Preprocessing Complexity

- The dataset contained continuous, binary, and Likert-scale ordinal variables.
- We explicitly converted ordinal variables to categorical types and standardized continuous features where appropriate. This ensured correct handling of mixed data and improved the stability of both dimensionality reduction and classification models.

Multicollinearity Among Predictors

- Certain features were found to be highly correlated, which can distort model interpretation and inflate variance.
- We assessed correlations and removed or combined redundant variables using feature selection techniques and model-based importance rankings.

Limited Sample Size for Deep Models

- The dataset size was insufficient for training high-capacity models like neural networks without overfitting.
- Focused on classical algorithms (Logistic Regression, Random Forest, XGBoost) and applied cross-validation to ensure robust performance.

Discussion and Conclusion

This study compared several machine learning approaches to predict student stress using psychological, physiological, academic, and social variables. The Random Forest classifier, trained on an optimal subset of features identified through ensemble feature selection, achieved the best performance with a test F1-score of 0.9045, demonstrating a robust ability to generalize, showing balanced performance across all three stress levels (No Stress, Moderate, High). While alternative approaches like Factor Analysis for Mixed Data (FAMD) were explored to uncover latent constructs, the direct use of carefully selected, interpretable features proved to be the most effective strategy for prediction.

The final Random Forest model provided high interpretability through feature importance and partial dependence analysis. Key insights from partial dependence plots revealed strong positive relationships between stress levels and factors like depression and anxiety, while academic_performance and teacher_student_relationship showed a clear negative association. Notably, the model highlighted the multifaceted nature of stress, with significant contributions spanning psychological well-being, social environment (e.g., bullying), academic pressures, and physiological factors (e.g., blood pressure).

Several analytical challenges were successfully addressed during the project. The presence of mixed data types (continuous, binary, and ordinal) was managed by leveraging algorithms like Random Forest that are inherently robust to such data, and by using FAMD for initial exploratory analysis. The issue of high dimensionality and potential overfitting was mitigated through a rigorous ensemble feature selection process, which distilled the original 20+ features down to an optimal set of 11.

In conclusion, the project developed a reliable, interpretable framework for predicting student stress. Findings emphasize that stress arises from complex interactions among multiple factors, with sleep quality and self-esteem as actionable targets. The Random Forest model offers a valuable tool for institutions to identify at-risk students and implement targeted support programs.

Appendix includes codes and technical details.

[EDA and Advance Analysis Codes](#)

References

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